

Birla Institute of Technology & Science, Pilani
Work-Integrated Learning Programmes Division
First Semester 2017-2018

Comprehensive Examination
(EC-3 Regular)

Course No. : SS ZG526
Course Title : Distributed Computing
Nature of Exam : Open Book
Weightage : 45%
Duration : 3 Hours
Date of Exam : 04/11/2017 (AN)

No. of Pages	= 4
No. of Questions	= 5

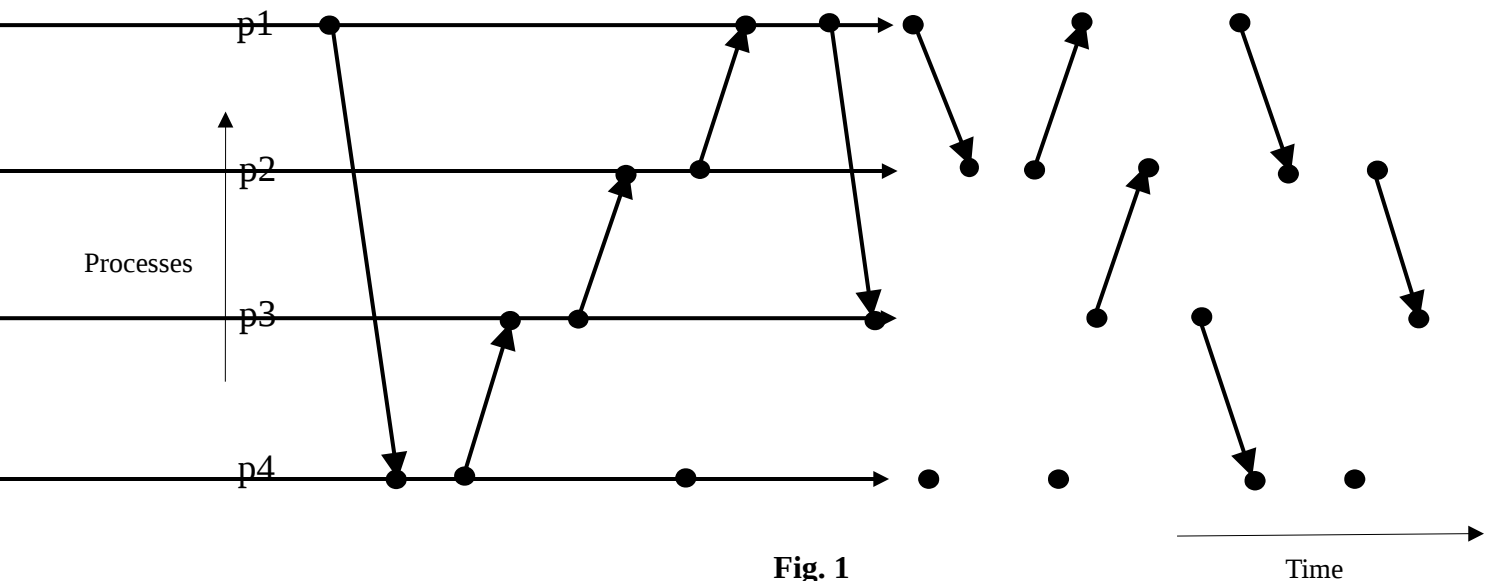
Note:

Please follow all the *Instructions to Candidates* given on the cover page of the answer book.

All parts of a question should be answered consecutively. Each answer should start from a fresh page.

Assumptions made if any, should be stated clearly at the beginning of your answer.

- Q.1 (a) Compute the scalar clock values of the events and the scalar timestamps of the messages exchanged among the 4 processes shown in the space-time diagram of Fig. 1. Assume that initially, the scalar clock values of all the processes is 0 and $d = 1$. [4]



- Q.1 (b) Compute the vector timestamps of the events occurring at the 4 processes and the vector timestamps of the messages exchanged among them as shown in the space-time diagram of Fig. 1. Use R1 and R2 for updating the vector clocks. Assume that initially, the vector clocks of each of the 4 processes is $[0\ 0\ 0\ 0]$ and $d = 1$. [5]
- Q.2 (a) Consider the space-time diagram of Fig. 2. Using Birman-Schiper-Stephenson (BSS) protocol, determine the progress of the vector clocks at each of the 3 processes corresponding to the occurrence of the different events, timestamps of the messages exchanged among the processes, conditions that need to be checked before message delivery and buffering of messages (if required). Note that message **a** is broadcast by p_2 in event e_2^1 and messages **b** and **c** are broadcast by p_3 in events e_3^3 and e_3^2 respectively. Assume that initially the vector clocks of all the 3 processes are $[0\ 0\ 0]$. [6]

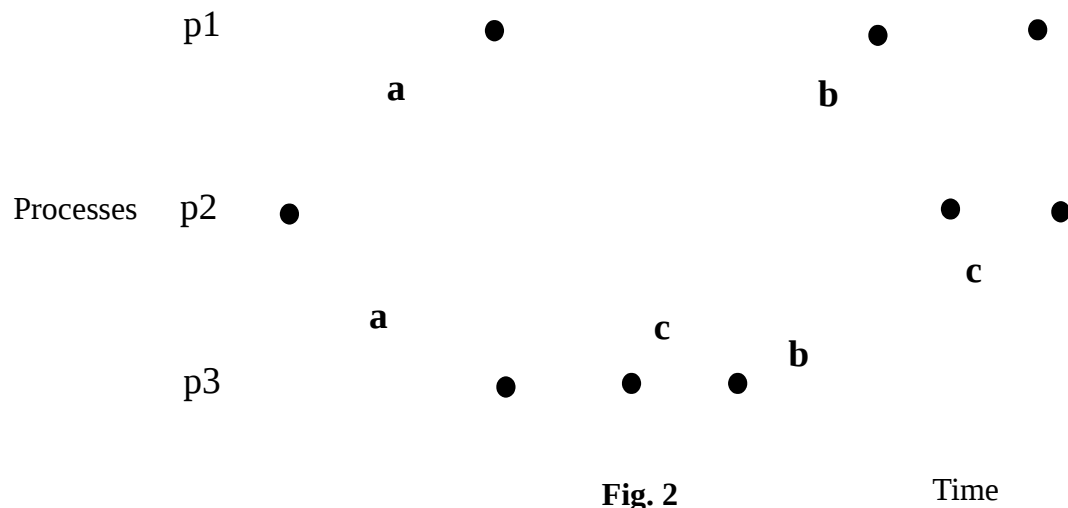


Fig. 2

- Q.2 (b) Apply synchronous single-initiator spanning tree algorithm using flooding on the graph shown in Fig. 3 to determine the BFS spanning tree and the round numbers in which the QUERY messages are sent by the different nodes. Assume A as the root node. [3]

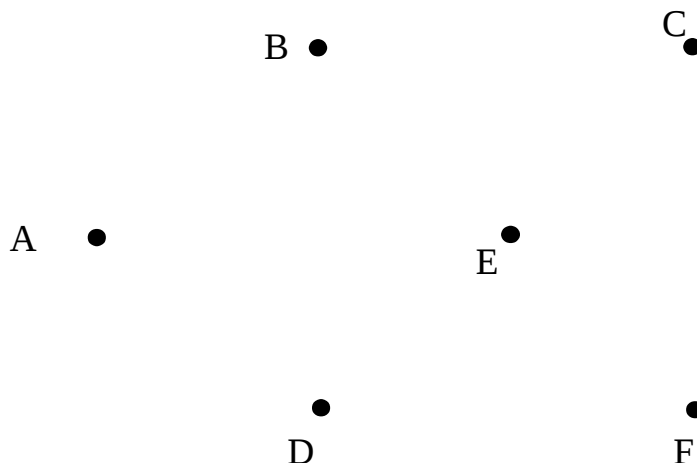


Fig. 3

- Q.3 (a) Consider a 3-stage Omega network consisting of 8 processors, 8 memory units and 12 switching elements each of size 2×2 . Between each pair of adjacent stages of the network, a link exists between output i of a stage and input j to the next stage according to the following interconnection function:

$$j = (2n - i) \% n$$

Note that for this network, $n = 8$ and $0 \leq i \leq 7$. Draw a diagram depicting the interconnections between the adjacent stages of this Omega network according to the above mentioned specifications. [5]

- Q.3 (b) Consider a distributed system consisting of 12 processes as depicted in Fig. 4.

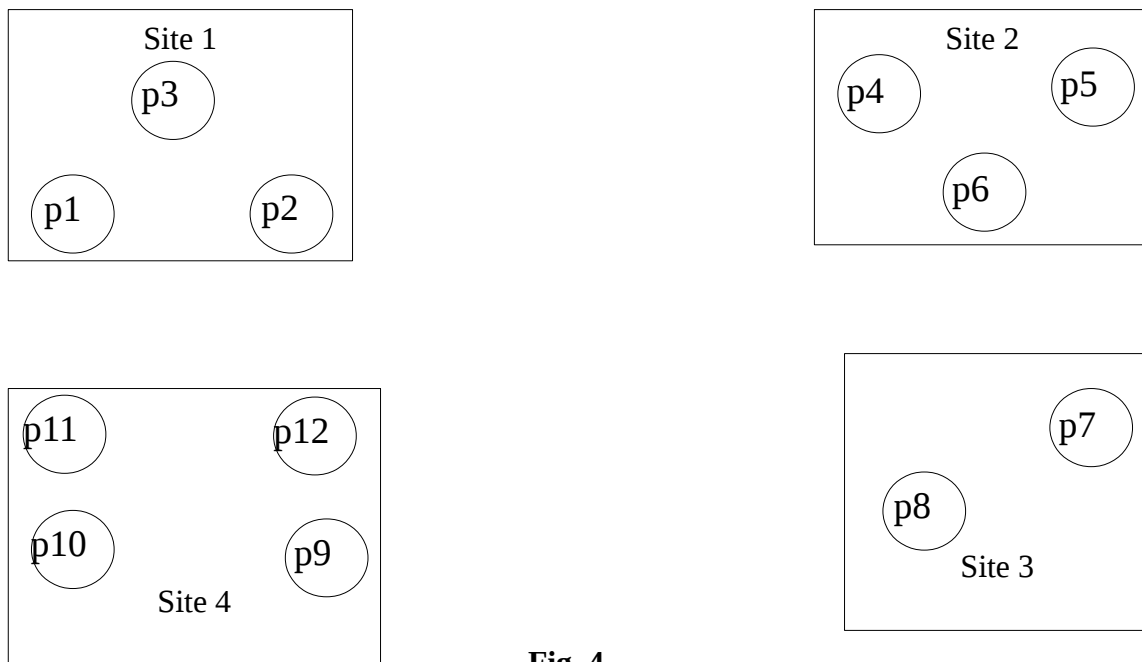


Fig. 4

The following dependencies exist among the processes:

- p_1 is blocked and is waiting for p_3 to release some resource
- p_2 is blocked and is waiting for p_3 to release some resource
- p_3 is blocked and is waiting for p_4 to release some resource
- p_4 is blocked and is waiting for p_5 to release some resource
- p_5 is blocked and is waiting for p_7 to release some resource
- p_7 is blocked and is waiting for p_8 to release some resource
- p_8 is blocked and is waiting for p_6 , p_9 and p_{12} to release some resources
- p_{12} is blocked and is waiting for p_{10} to release some resource
- p_{10} is blocked and is waiting for p_{11} to release some resource
- p_{11} is blocked and is waiting for p_2 to release some resource

- i. Draw the WFG for the above scenario.

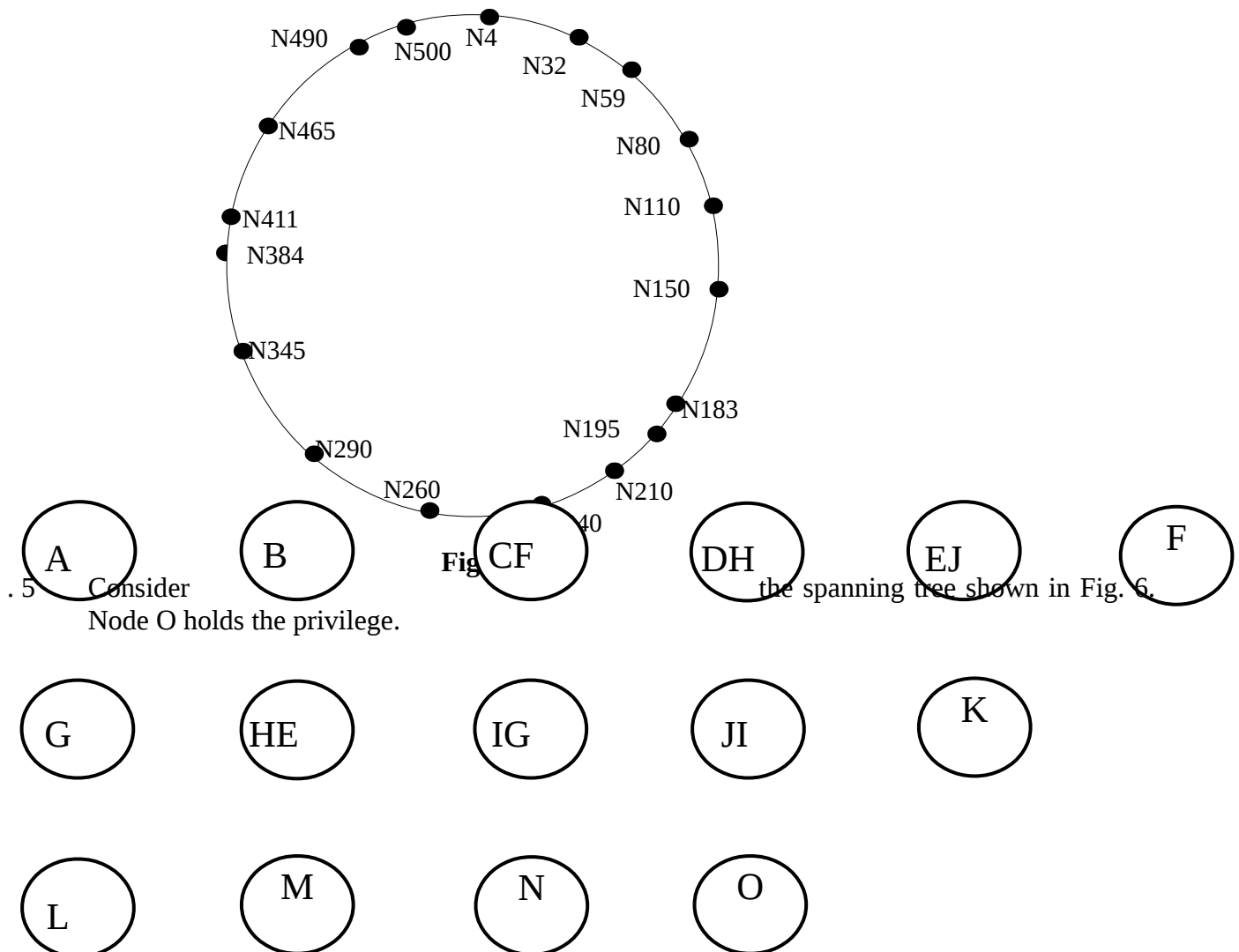
- ii. Assuming that the WFG determined in (i) is an AND model WFG, determine whether a deadlock exists in the system with proper reason. If a deadlock exists, identify the deadlocked processes. [2 + 2 = 4]

Q.4 (a) A distributed system consists of 12 sites - $S_1, S_2, S_3, S_4, S_5, S_6, S_7, S_8, S_9, S_{10}, S_{11}$ and S_{12} . The request sets for S_4, S_5 and S_6 are R_4, R_5 and R_6 respectively.

$R_4 = \{S_1, S_2, S_4, S_7, S_{10}\}, R_5 = \{S_3, S_5, S_7, S_8, S_9\}, R_6 = \{S_6, S_8, S_{10}, S_{11}, S_{12}\}$

Give an example scenario which results in a deadlock when sites S_4, S_5 and S_6 simultaneously invoke mutual exclusion using Maekawa's algorithm. Clearly mention the sites locked by each of S_4, S_5 and S_6 and the sites for which each of S_4, S_5 and S_6 waits. [3]

Q.4 (b) Fig. 5 shows the arrangement of nodes along a Chord ring with $m = 9$. Determine the finger tables for N_{32}, N_{183} and N_{240} . [2 + 2 + 2 = 6]



5. Consider the spanning tree shown in Fig. 6. Node O holds the privilege.

Using the knowledge of Raymond's tree-based algorithm, answer the following questions.

- a) Determine the values of the HOLDER variables of each of the nodes. [8]
- b) Redraw the spanning tree of Fig. 6 by replacing the undirected edges by directed edges. [1]